Lab Assignment 5: Interactive Visual Analytics using Tableau

Objective: Design interactive dashboards and stories using Tableau.
Dataset: Sample - Superstore (`Sample - Superstore.csv`, 9,994 order lines, 21 columns, orders from Jan 2015 to Dec 2018, United States).
Tool: Tableau Desktop or Tableau Public (free). The same steps work in both.
Deliverables: 4 worksheets (bar, line, map, scatter), one interactive dashboard with filter and highlight actions, and a story.

Note: Tableau is a point-and-click tool, so this lab is done by following the procedure in Sections 1-5. Section 6 lists the exact values each sheet should show and Section 7 has reference previews made in Python from the same CSV file (code in the Appendix), so you can check your results. For your submission, add screenshots of your own Tableau sheets and dashboard (*Worksheet > Export > Image* or *Dashboard > Export Image*).

1. Import the Dataset and Inspect Data Types

1. Open Tableau. On the start page, under **Connect > To a File**, click **Text file** and choose `Sample - Superstore.csv`.
2. The **Data Source** page opens and shows the first rows. The icon above each column header is its data type: **Abc** = string, **#** = number, **calendar** = date, **globe** = geographic role.
3. Check that **Order Date** and **Ship Date** show the date icon. If they show **Abc**, click the icon and choose **Date**.
4. Click the icon on **Postal Code**: change it to **String**, then **Geographic Role > ZIP Code/Postcode**. ZIP codes are labels, not numbers to add up.
5. Check that **Country**, **State** and **City** show the globe icon (set **Geographic Role** if not).
6. Click **Sheet 1** at the bottom left. In the **Data** pane, the fields are split into **Dimensions** (categories, blue) and **Measures** (numbers, green).

Table 1: Fields and their Tableau data types

Field	Tableau data type	Role	Distinct	Notes
Row ID	# Number (whole)	Dimension	9,994	Line number, not a measure
Order ID	Abc String	Dimension	5,009	One order has many lines
Order Date, Ship Date	Date	Dimension	1,236 / 1,334	03-Jan-2015 to 30-Dec-2018
Ship Mode	Abc String	Dimension	4	Standard, Second, First Class, Same Day
Customer ID, Customer Name	Abc String	Dimension	793	
Segment	Abc String	Dimension	3	Consumer, Corporate, Home Office
Country	Abc String (geo: Country)	Dimension	1	United States
State	Abc String (geo: State)	Dimension	49	
City	Abc String (geo: City)	Dimension	531	
Postal Code	Abc String (geo: ZIP Code)	Dimension	630	11 blanks (Burlington, VT)
Region	Abc String	Dimension	4	Central, East, South, West
Product ID, Product Name	Abc String	Dimension	1,862 / 1,850	
Category, Sub-Category	Abc String	Dimension	3 / 17	
Sales	# Number (decimal)	Measure		Total $2,297,200.86
Quantity	# Number (whole)	Measure		
Discount	# Number (decimal)	Measure		0 to 0.8
Profit	# Number (decimal)	Measure		Total $286,397.02 (can be negative)

2. Build the Individual Worksheets

Worksheet 1 - Bar chart: Sales and Profit by Region

1. Double-click the **Sheet 1** tab and rename it **Sales by Region**.
2. Drag **Region** to the **Columns** shelf.
3. Drag **Sales** to the **Rows** shelf, then drag **Profit** to **Rows** next to it. Tableau draws two bar panes, one for SUM(Sales) and one for SUM(Profit).
4. On the **Marks** card (**All** tab), make sure the mark type is **Bar**. Drag **Region** to **Color**.
5. Click **Label** and tick **Show mark labels**. To show currency, right-click SUM(Sales) > **Format** > Numbers > Currency (Custom), 0 decimals, units Thousands (K).
6. Sort the regions by sales: hover over the Sales axis and click the sort icon (descending).

Worksheet 2 - Line chart: Sales trend by month

1. Click the **New Worksheet** icon and rename it **Monthly Sales Trend**.
2. Drag **Order Date** to **Columns**. Tableau shows YEAR(Order Date).
3. Click the drop-down arrow on the YEAR(Order Date) pill and choose the **second Month** option (the one with an example like *May 2015*). The pill turns green: a continuous monthly time axis.
4. Drag **Sales** to **Rows**. The mark type changes to **Line** automatically.
5. Drag **Profit** to **Tooltip** on the Marks card so hovering a point shows both values.
6. Optional: open the **Analytics** pane and drag **Trend Line** onto the view (**Linear**) to highlight the overall upward trend.
7. Optional (year-over-year view): use the first (discrete) **Month** option on Columns and drag **Order Date** to **Color** as YEAR to get one line per year.

Worksheet 3 - Map: Sales by State (geographic breakdown)

1. New worksheet, rename it **Sales Map**.
2. Double-click **State**. Tableau builds a map using the generated Latitude and Longitude.
3. Drag **Sales** to **Color**. Change the mark type to **Map** for a filled map (darker = more sales).
4. Drag **Profit** to **Tooltip**, and drag **Region** to **Detail** (Region is needed for the highlight action later).
5. Optional: put **Profit** on Color instead and choose the **Red-Green Diverging** palette (Color > Edit Colors) to show loss-making states in red.
6. If a state shows as unknown, use **Map > Edit Locations** and set Country to United States.

Worksheet 4 - Scatter plot: Customer Sales vs Profit by Segment

1. New worksheet, rename it **Customer Segments**.
2. Drag **Sales** to **Columns** and **Profit** to **Rows**. You see a single mark (the totals).
3. Drag **Customer Name** to **Detail** on the Marks card. You now get one mark per customer: 793 points.
4. Drag **Segment** to **Color** and set the mark type to **Circle**.
5. Optional: Analytics pane > drag **Trend Line** onto the view to show the Sales-Profit relationship per segment.

3. Combine the Sheets into an Interactive Dashboard

1. Click the **New Dashboard** icon at the bottom (or menu **Dashboard > New Dashboard**). Rename it **Superstore Sales Dashboard**.
2. Under **Size** on the left, choose **Fixed size** 1200 x 800 (or **Automatic**).
3. Keep **Tiled** selected and drag the sheets from the **Sheets** list onto the canvas: **Sales by Region** top-left, **Customer Segments** top-right, **Sales Map** bottom-left and **Monthly Sales Trend** bottom-right (or across the bottom).
4. Tick **Show dashboard title** at the bottom left and edit the title text.
5. Optional global filter: on any sheet in the dashboard, click its drop-down arrow > **Filters** > **Category**. Then on the new filter card choose **Apply to Worksheets > All Using This Data Source**.
6. Remove or float any extra legends you don't need, and resize tiles so every chart is readable.

4. Add Interactive Dashboard Actions (Filter / Highlight)

Open **Dashboard > Actions...** and use **Add Action** for each action below.

Action	Type	Source sheet	Run action on	Target sheets	Settings
Filter by Region	Filter	Sales by Region	Select	Monthly Sales Trend, Sales Map, Customer Segments	Clearing the selection will: **Show all values**; Filter: All Fields
Filter by State	Filter	Sales Map	Select	Monthly Sales Trend, Customer Segments	Clearing the selection will: **Show all values**
Highlight Region	Highlight	Sales Map	Hover	Sales by Region	Target Highlighting: **Selected Fields > Region**

1. Shortcut for a filter action: select a sheet on the dashboard and click its funnel icon (**Use as Filter**). Tableau creates a Select filter action for you.
2. For a quick highlight by segment, click the highlighter icon on the **Segment** colour legend. Clicking Consumer, Corporate or Home Office then highlights those customers.
3. **Test it:** click the **West** bar. The map shows only the 11 West states, the trend line shows West monthly sales and the scatter shows the 686 West customers. Click the bar again (or press Esc) to clear the filter.
4. Hover over a state on the map: the bar for that state's region is highlighted.

5. Create a Story and Save / Publish

1. Click **New Story** at the bottom. Drag **Superstore Sales Dashboard** into the story and type a caption such as *Overview: $2.30M sales, $286K profit*.
2. Click **Duplicate** to add a new story point, click the **West** bar, and caption it *West leads with $725K sales and a 14.9% profit ratio*.
3. Add more points, e.g. *Texas, Ohio and Pennsylvania lose money* (Sales Map) and *Sales peak every Sep, Nov and Dec* (Monthly Sales Trend).
4. Save: Tableau Desktop - **File > Save As** a packaged workbook (.twbx, includes the data). Tableau Public - **File > Save to Tableau Public As...**, then share the link.

6. Expected Results (values to check in Tableau)

Table 2: Worksheet 1 - SUM(Sales) and SUM(Profit) by Region

Region	Sales	Profit	Profit ratio
West	$725,457.82	$108,418.45	14.9%
East	$678,781.24	$91,522.78	13.5%
Central	$501,239.89	$39,706.36	7.9%
South	$391,721.90	$46,749.43	11.9%
Total	**$2,297,200.86**	**$286,397.02**	**12.5%**

Table 3: Worksheet 2 - Sales and Profit by Year

Year	Sales	Profit	Sales growth
2015	$484,247.50	$49,543.97	-
2016	$470,532.51	$61,618.60	-2.8%
2017	$609,205.60	$81,795.17	+29.5%
2018	$733,215.26	$93,439.27	+20.4%

Highest month: Nov 2018 ($118,447.82). Lowest month: Feb 2015 ($4,519.89).

Table 4: Worksheets 3 and 4 - State and Segment breakdowns

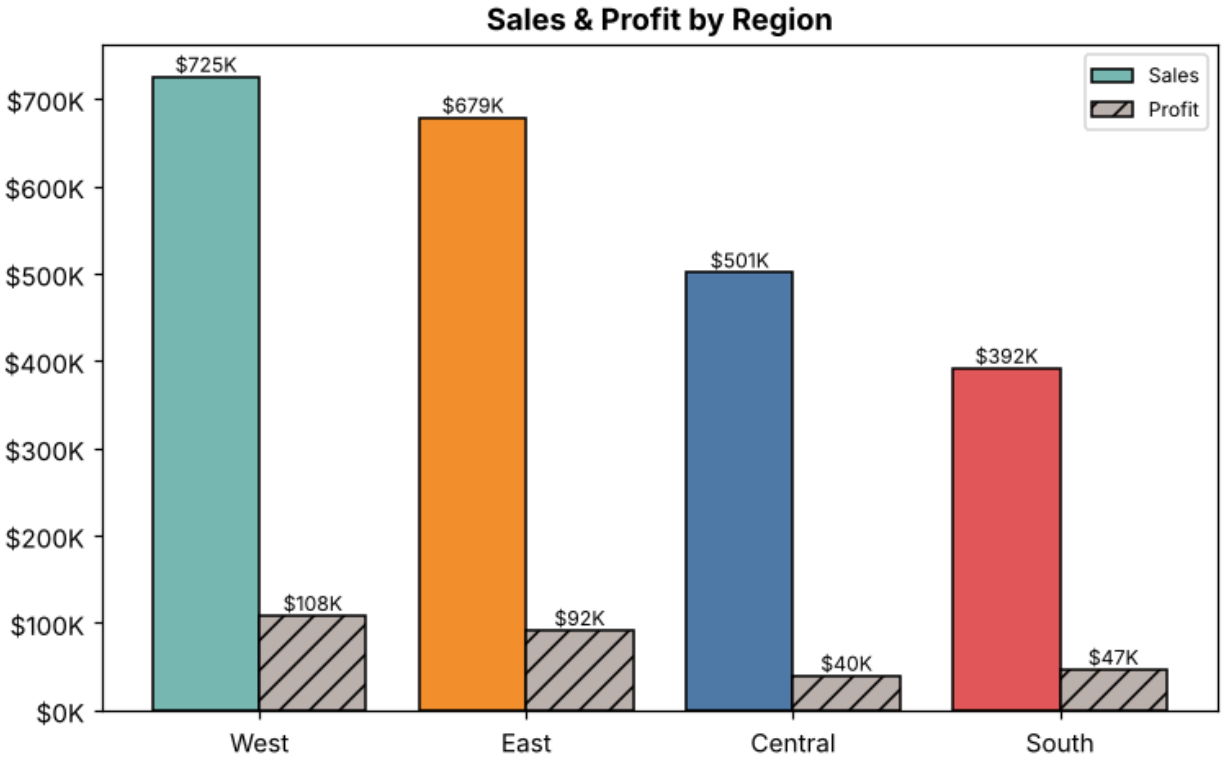
Top 5 states by Sales	Sales	Profit
California	$457,687.63	$76,381.39
New York	$310,876.27	$74,038.55
Texas	$170,188.05	-$25,729.36
Washington	$138,641.27	$33,402.65
Pennsylvania	$116,511.91	-$15,559.96

Segment	Sales	Profit	Customers
Consumer	$1,161,401.34	$134,119.21	409
Corporate	$706,146.37	$91,979.13	236
Home Office	$429,653.15	$60,298.68	148

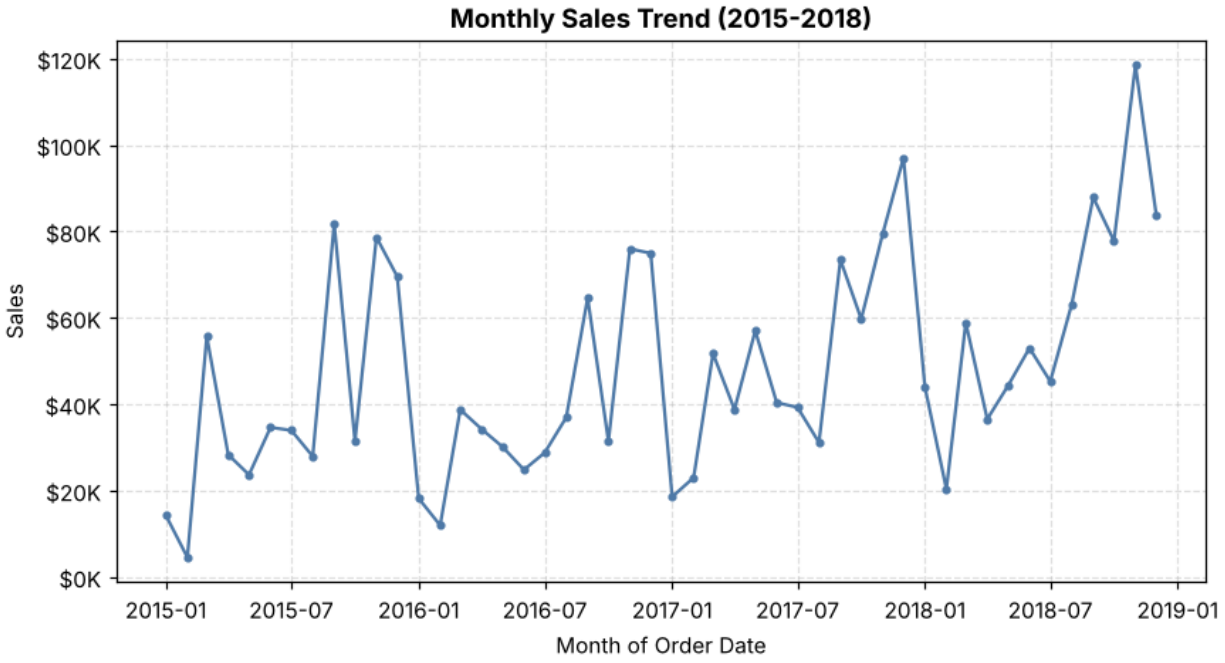
After clicking Region = West (filter action): 3,203 order lines, Sales $725,457.82, Profit $108,418.45, 686 customers, 11 states.

7. Reference Previews (made in Python from the same data)

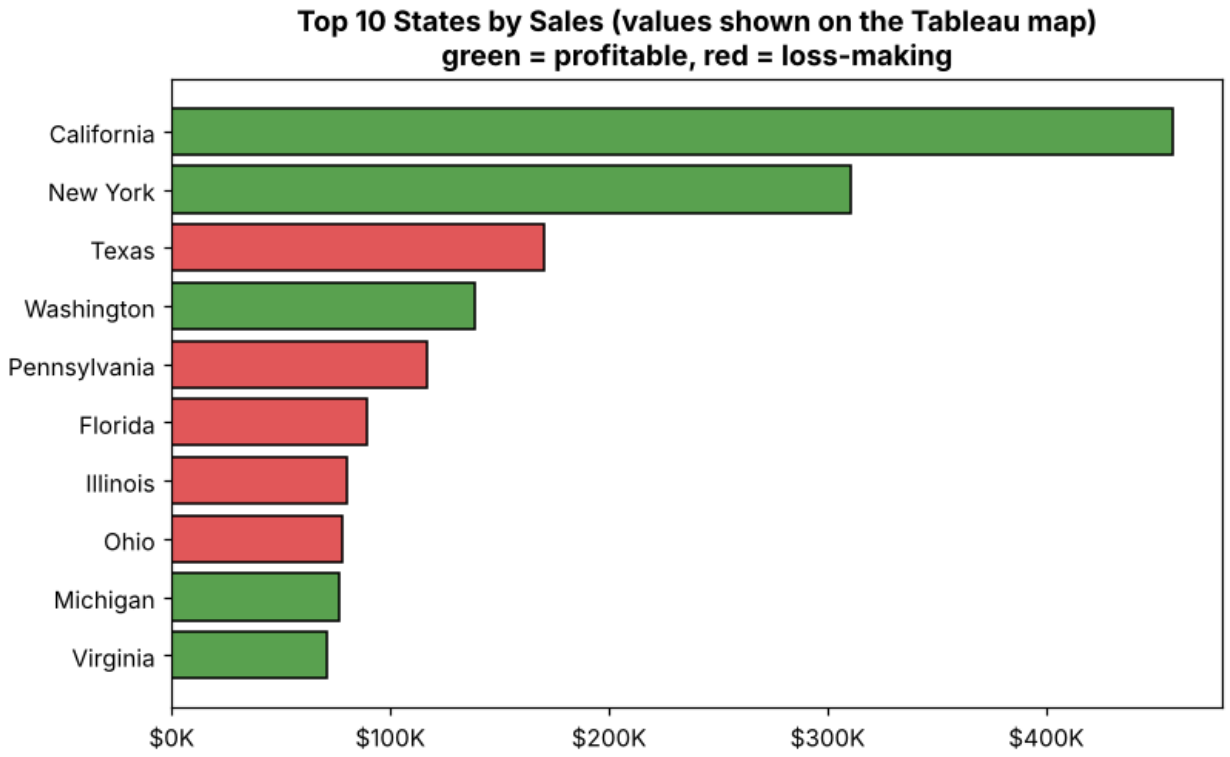
Preview 1: Worksheet 1 - Sales & Profit by Region



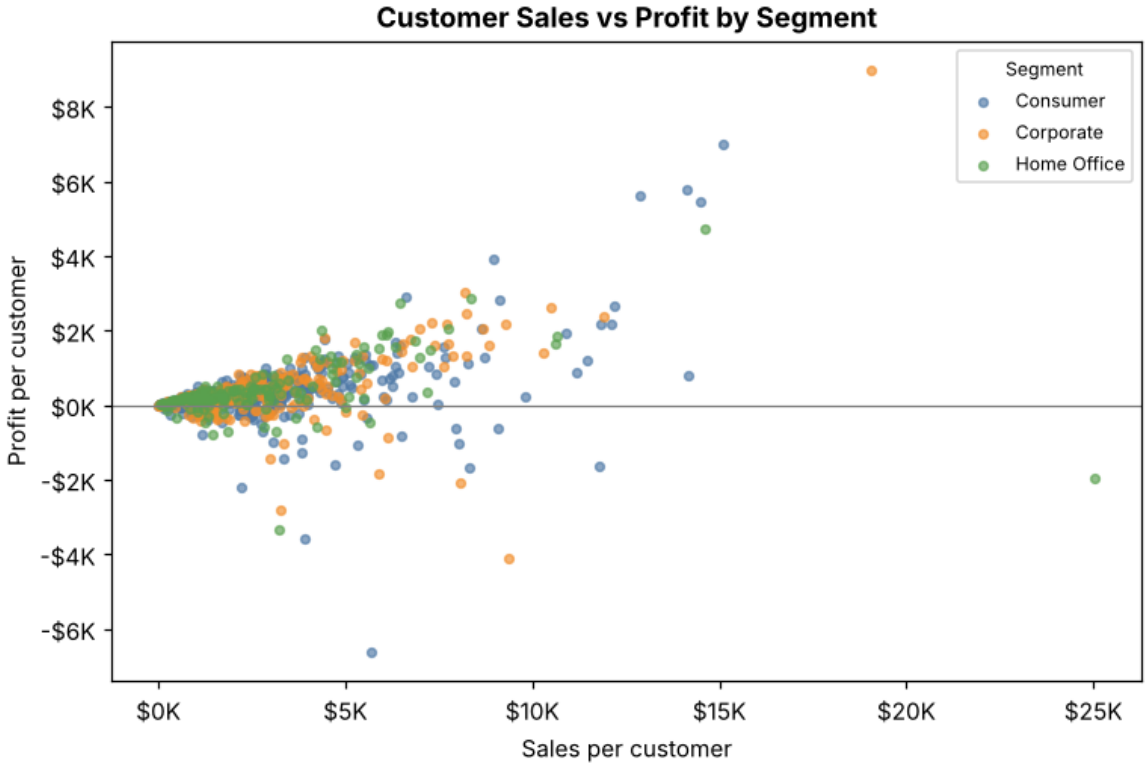
Preview 2: Worksheet 2 - Monthly Sales Trend



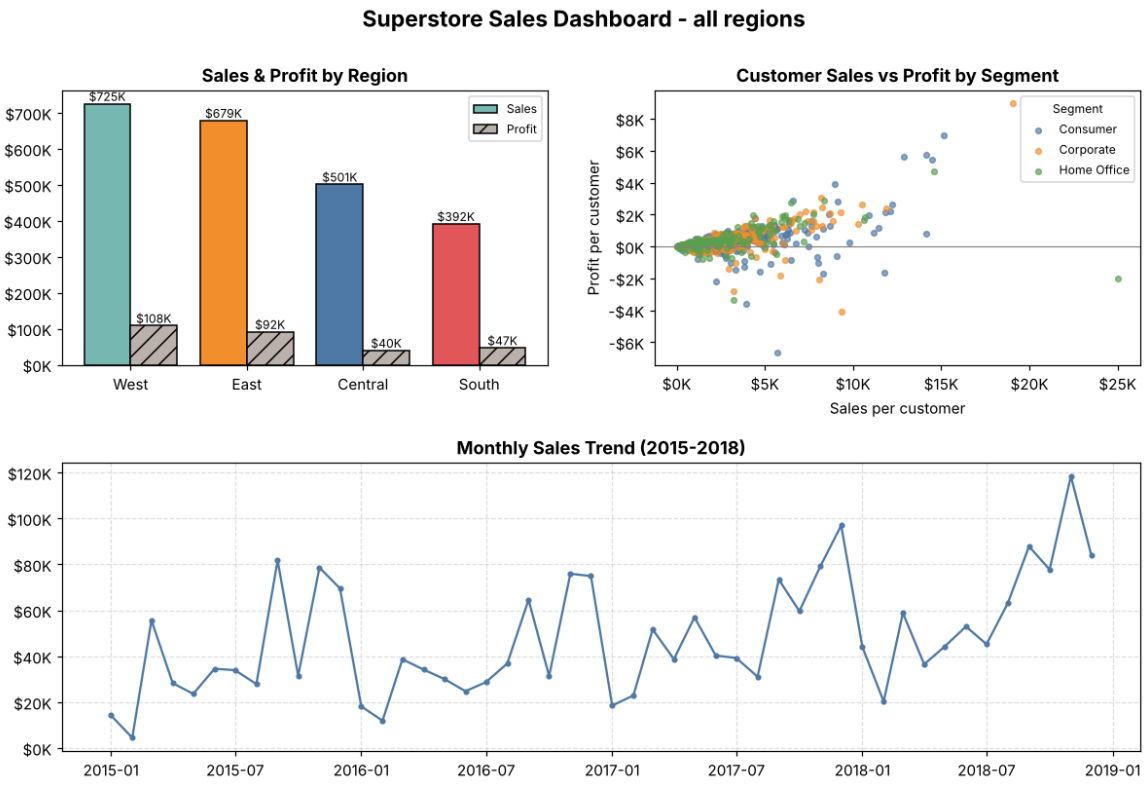
Preview 3: Worksheet 3 - Top 10 states by Sales (the values on the map)



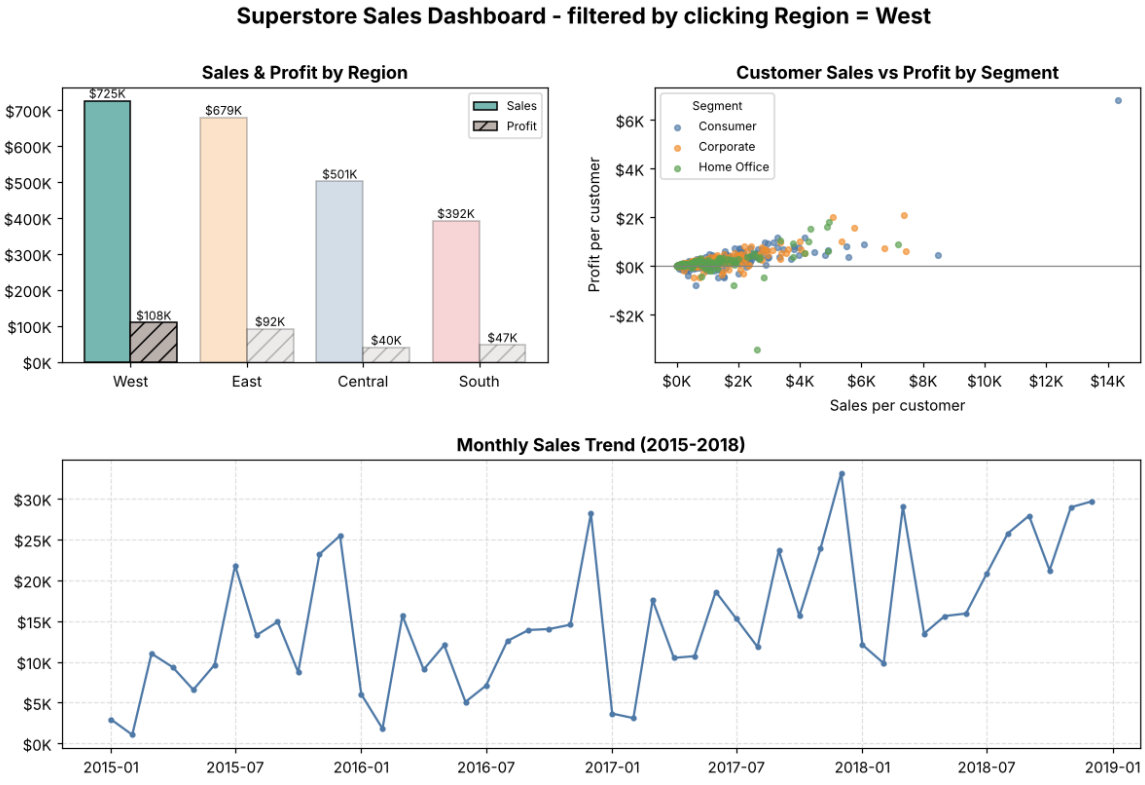
Preview 4: Worksheet 4 - Customer Sales vs Profit by Segment



Preview 5: Dashboard layout - all regions



Preview 6: Dashboard after clicking the West bar (filter action applied to the other sheets)



Appendix A: Python Companion Code (data checks and previews)

```
# ======================================================================
# Lab Assignment 5: Interactive Visual Analytics using Tableau
#                   -- Python companion script --
# ----------------------------------------------------------------------
# The dashboard itself is built in Tableau (see the PDF for the steps).
# This script inspects the same "Sample - Superstore.csv" file and
# computes every number each Tableau worksheet should show, so the
# Tableau results can be checked. It also draws preview images of the
# worksheets and of the dashboard before / after the Region filter.
#
# Dataset   : Sample - Superstore.csv (9,994 order lines, 2015-2018)
# Run with  : python lab5_superstore_dashboard_preview.py
# Requires  : pip install pandas matplotlib
# Output    : Step-by-step console log + PNG previews in ./plots/
# ======================================================================

import os
from datetime import datetime

import matplotlib
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import pandas as pd


# ---------- Helper: print a clearly visible step header --------------
def step(n, title):
    print("\n" + "=" * 70)
    print(f"STEP {n} : {title}")
    print("=" * 70)


OUT_DIR = "plots"
os.makedirs(OUT_DIR, exist_ok=True)


def save_plot(fig, file):
    path = os.path.join(OUT_DIR, file)
    fig.savefig(path, dpi=110, bbox_inches="tight")
    print(">> Plot saved to:", path)
    if matplotlib.get_backend().lower() != "agg":
        plt.show()
    plt.close(fig)


money = mticker.FuncFormatter(lambda v, _: f"{'-' if v < 0 else ''}${abs(v) / 1000:,.0f}K")
REGION_COLORS = {"Central": "#4e79a7", "East": "#f28e2b", "South": "#e15759", "West": "#76b7b2"}
SEGMENT_COLORS = {"Consumer": "#4e79a7", "Corporate": "#f28e2b", "Home Office": "#59a14f"}

pd.set_option("display.width", 110)
pd.set_option("display.max_columns", None)
print("Lab Assignment 5 - Tableau Superstore Dashboard (Python companion)")
print("Python libraries : pandas", pd.__version__, "| matplotlib", matplotlib.__version__)
print("Run on           :", datetime.now().strftime("%Y-%m-%d %H:%M:%S"))


# ======================================================================
# STEP 1 : Import the dataset and inspect data types
# ======================================================================
step(1, "Import 'Sample - Superstore.csv' and inspect data types")

df = pd.read_csv("Sample - Superstore.csv", parse_dates=["Order Date", "Ship Date"])
print(">> Dataset loaded successfully.")
print(">> Shape (rows x columns):", df.shape)

# How Tableau will classify each column on its Data Source page
def tableau_type(col):
    if col in ("Country", "State", "City"):
        return "Abc  String (geographic role)", "Dimension"
    if col == "Postal Code":
        return "Abc  String (set geo role: ZIP Code)", "Dimension"
    if pd.api.types.is_datetime64_any_dtype(df[col]):
        return "Date", "Dimension"
    if col == "Row ID":
```

```
        return "#    Number (whole)", "Dimension"
    if col == "Quantity":
        return "#    Number (whole)", "Measure"
    if pd.api.types.is_numeric_dtype(df[col]):
        return "#    Number (decimal)", "Measure"
    return "Abc  String", "Dimension"

types = pd.DataFrame({
    "pandas dtype": df.dtypes.astype(str),
    "Tableau data type": [tableau_type(c)[0] for c in df.columns],
    "Tableau role": [tableau_type(c)[1] for c in df.columns],
    "unique values": df.nunique(),
})
print("\n>> Column data types (pandas vs Tableau):")
print(types)

print("\n>> Missing values (non-zero only):")
print(df.isnull().sum()[lambda s: s > 0])
print(f"\n>> Order Date range : {df['Order Date'].min():%d-%b-%Y} to "
      f"{df['Order Date'].max():%d-%b-%Y}")
print(f">> Total Sales      : ${df['Sales'].sum():,.2f}")
print(f">> Total Profit     : ${df['Profit'].sum():,.2f}")
print(f">> Profit ratio     : {df['Profit'].sum() / df['Sales'].sum():.2%}")


# ======================================================================
# STEP 2 : Worksheet 1 - Bar chart of Sales and Profit by Region
# ======================================================================
step(2, "Worksheet 1 - Bar chart: total Sales and Profit by Region")

region = (df.groupby("Region")[["Sales", "Profit"]].sum()
            .sort_values("Sales", ascending=False))
region["Profit Ratio %"] = region["Profit"] / region["Sales"] * 100
print(">> SUM(Sales) and SUM(Profit) per Region (what Tableau should show):")
print(region.round(2))


def draw_region_bars(ax, data, selected=None):
    # selected = region clicked on the dashboard; other bars are faded
    x = range(len(data))
    alpha = [1.0 if selected in (None, r) else 0.25 for r in data.index]
    for i, r in enumerate(data.index):
        ax.bar(i - 0.2, data.loc[r, "Sales"], width=0.4, label="Sales" if i == 0 else None,
               color=REGION_COLORS[r], edgecolor="black", alpha=alpha[i])
        ax.bar(i + 0.2, data.loc[r, "Profit"], width=0.4, label="Profit" if i == 0 else None,
               color="#bab0ac", edgecolor="black", hatch="//", alpha=alpha[i])
    for i, (s, p) in enumerate(zip(data["Sales"], data["Profit"])):
        ax.text(i - 0.2, s, f"${s / 1000:,.0f}K", ha="center", va="bottom", fontsize=8)
        ax.text(i + 0.2, p, f"${p / 1000:,.0f}K", ha="center", va="bottom", fontsize=8)
    ax.set_xticks(list(x))
    ax.set_xticklabels(data.index)
    ax.yaxis.set_major_formatter(money)
    ax.set_title("Sales & Profit by Region", fontweight="bold")
    ax.legend(loc="upper right", fontsize=8)


fig, ax = plt.subplots(figsize=(8, 4.8))
draw_region_bars(ax, region)
save_plot(fig, "1_sheet_region_bar.png")
print(f">> Interpretation: {region.index[0]} has the highest sales and profit; "
      f"{region['Profit Ratio %'].idxmin()} has the lowest profit ratio "
      f"({region['Profit Ratio %'].min():.1f}%).")


# ======================================================================
# STEP 3 : Worksheet 2 - Line chart of monthly Sales (time series)
# ======================================================================
step(3, "Worksheet 2 - Line chart: Sales by month and year")

yearly = df.groupby(df["Order Date"].dt.year)[["Sales", "Profit"]].sum()
yearly.index.name = "Year"
yearly["Sales growth %"] = yearly["Sales"].pct_change() * 100
print(">> SUM(Sales) and SUM(Profit) per YEAR(Order Date):")
print(yearly.round(2))

monthly = df.set_index("Order Date")["Sales"].resample("MS").sum()
```

```
print("\n>> Continuous MONTH(Order Date) series: 48 months; peaks and lows:")
print(f"   Highest month : {monthly.idxmax():%b %Y} (${monthly.max():,.2f})")
print(f"   Lowest month  : {monthly.idxmin():%b %Y} (${monthly.min():,.2f})")

by_month = (df.assign(Year=df["Order Date"].dt.year, Month=df["Order Date"].dt.month)
              .pivot_table(values="Sales", index="Month", columns="Year", aggfunc="sum"))
print("\n>> Sales by month (rows) and year (columns), in USD:")
print(by_month.round(0))


def draw_monthly_line(ax, series):
    ax.plot(series.index, series.values, color="#4e79a7", marker="o", markersize=3)
    ax.yaxis.set_major_formatter(money)
    ax.set_title("Monthly Sales Trend (2015-2018)", fontweight="bold")
    ax.grid(True, linestyle="--", alpha=0.4)


fig, ax = plt.subplots(figsize=(9, 4.5))
draw_monthly_line(ax, monthly)
ax.set_xlabel("Month of Order Date")
ax.set_ylabel("Sales")
save_plot(fig, "2_sheet_monthly_sales_line.png")
print(">> Interpretation: sales dipped 2.8% in 2016, then grew strongly in 2017 and 2018.\n"
      "   Clear seasonal pattern: low in Jan/Feb, highest in Sep, Nov and Dec.")


# ======================================================================
# STEP 4 : Worksheet 3 - Map (Sales by State) and segment scatter
# ======================================================================
step(4, "Worksheet 3 - Geographic (State map) and Segment (scatter) breakdowns")

state = df.groupby("State")[["Sales", "Profit"]].sum().sort_values("Sales", ascending=False)
print(f">> Filled map values: {len(state)} states. Top 10 states by SUM(Sales):")
print(state.head(10).round(2))
print("\n>> 5 states with the largest losses (shown red if Profit is on Color):")
print(state.sort_values("Profit").head(5).round(2))

segment = df.groupby("Segment")[["Sales", "Profit"]].sum()
segment["Customers"] = df.groupby("Segment")["Customer ID"].nunique()
print("\n>> Segment breakdown:")
print(segment.round(2))

customers = df.groupby(["Customer Name", "Segment"])[["Sales", "Profit"]].sum().reset_index()
print(f"\n>> Scatter plot: one mark per customer ({len(customers)} marks),"
      " x = SUM(Sales), y = SUM(Profit), color = Segment")

fig, ax = plt.subplots(figsize=(8, 4.8))
top10 = state.head(10)[::-1]
ax.barh(top10.index, top10["Sales"],
        color=["#59a14f" if p > 0 else "#e15759" for p in top10["Profit"]], edgecolor="black")
ax.xaxis.set_major_formatter(money)
ax.set_title("Top 10 States by Sales (values shown on the Tableau map)\n"
             "green = profitable, red = loss-making", fontweight="bold")
save_plot(fig, "3_sheet_state_sales_top10.png")


def draw_segment_scatter(ax, data):
    for seg, g in data.groupby("Segment"):
        ax.scatter(g["Sales"], g["Profit"], s=14, alpha=0.65,
                   color=SEGMENT_COLORS[seg], label=seg)
    ax.axhline(0, color="grey", linewidth=0.8)
    ax.xaxis.set_major_formatter(money)
    ax.yaxis.set_major_formatter(money)
    ax.set_xlabel("Sales per customer")
    ax.set_ylabel("Profit per customer")
    ax.set_title("Customer Sales vs Profit by Segment", fontweight="bold")
    ax.legend(fontsize=8, title="Segment", title_fontsize=8)


fig, ax = plt.subplots(figsize=(8, 5))
draw_segment_scatter(ax, customers)
save_plot(fig, "4_sheet_segment_scatter.png")
print(f">> Interpretation: {state.index[0]} and {state.index[1]} lead sales; "
      f"{state['Profit'].idxmin()} loses the most money.")


```

```
# ======================================================================
# STEP 5 : Dashboard preview and the Region filter action
# ======================================================================
step(5, "Dashboard preview - all regions vs Region = 'West' selected")


def draw_dashboard(data, title, file, selected=None):
    # The source sheet (Region bars) keeps all regions; only the targets are filtered
    fig = plt.figure(figsize=(13, 8))
    grid = fig.add_gridspec(2, 2, height_ratios=[1, 1.05], hspace=0.35, wspace=0.22)
    draw_region_bars(fig.add_subplot(grid[0, 0]), region, selected)
    draw_segment_scatter(fig.add_subplot(grid[0, 1]),
                         data.groupby(["Customer Name", "Segment"])[["Sales", "Profit"]]
                             .sum().reset_index())
    ax = fig.add_subplot(grid[1, :])
    draw_monthly_line(ax, data.set_index("Order Date")["Sales"].resample("MS").sum())
    fig.suptitle(title, fontsize=15, fontweight="bold")
    save_plot(fig, file)


draw_dashboard(df, "Superstore Sales Dashboard - all regions", "5_dashboard_all_regions.png")

west = df[df["Region"] == "West"]
print("\n>> After clicking the 'West' bar (Filter action), the other sheets show:")
print(f"   Order lines       : {len(west):,}")
print(f"   Total Sales       : ${west['Sales'].sum():,.2f}")
print(f"   Total Profit      : ${west['Profit'].sum():,.2f}")
print(f"   Customers in West : {west['Customer ID'].nunique()}")
print(f"   States in West    : {west['State'].nunique()}")
draw_dashboard(west, "Superstore Sales Dashboard - filtered by clicking Region = West",
               "6_dashboard_filtered_west.png", selected="West")


# ======================================================================
# Done
# ======================================================================
step(6, "Lab Assignment 5 companion script completed")
print("Files generated:")
for f in sorted(os.listdir(OUT_DIR)):
    print("  -", os.path.join(OUT_DIR, f))
```

Appendix B: Execution Output (console printout)

```
Lab Assignment 5 - Tableau Superstore Dashboard (Python companion)
Python libraries : pandas 3.0.5 | matplotlib 3.11.2
Run on           : 2026-10-26 21:35:07

======================================================================
STEP 1 : Import 'Sample - Superstore.csv' and inspect data types
======================================================================
>> Dataset loaded successfully.
>> Shape (rows x columns): (9994, 21)

>> Column data types (pandas vs Tableau):
                pandas dtype                    Tableau data type Tableau role  unique values
Row ID                 int64                  #    Number (whole)    Dimension           9994
Order ID                 str                       Abc  String    Dimension           5009
Order Date     datetime64[us]                         Date    Dimension           1236
Ship Date      datetime64[us]                         Date    Dimension           1334
Ship Mode                str                       Abc  String    Dimension              4
Customer ID              str                       Abc  String    Dimension            793
Customer Name            str                       Abc  String    Dimension            793
Segment                  str                       Abc  String    Dimension              3
Country                  str         Abc  String (geographic role)    Dimension              1
City                     str         Abc  String (geographic role)    Dimension            531
State                    str         Abc  String (geographic role)    Dimension             49
Postal Code          float64  Abc  String (set geo role: ZIP Code)    Dimension            630
Region                   str                       Abc  String    Dimension              4
Product ID               str                       Abc  String    Dimension           1862
Category                 str                       Abc  String    Dimension              3
Sub-Category             str                       Abc  String    Dimension             17
Product Name             str                       Abc  String    Dimension           1850
Sales                float64                #    Number (decimal)      Measure           5825
Quantity               int64                  #    Number (whole)      Measure             14
Discount             float64                #    Number (decimal)      Measure             12
Profit               float64                #    Number (decimal)      Measure           7287

>> Missing values (non-zero only):
Postal Code    11
dtype: int64

>> Order Date range : 03-Jan-2015 to 30-Dec-2018
>> Total Sales      : $2,297,200.86
>> Total Profit     : $286,397.02
>> Profit ratio     : 12.47%

======================================================================
STEP 2 : Worksheet 1 - Bar chart: total Sales and Profit by Region
======================================================================
>> SUM(Sales) and SUM(Profit) per Region (what Tableau should show):
             Sales     Profit  Profit Ratio %
Region
West     725457.82  108418.45           14.94
East     678781.24   91522.78           13.48
Central  501239.89   39706.36            7.92
South    391721.90   46749.43           11.93
>> Plot saved to: plots/1_sheet_region_bar.png
>> Interpretation: West has the highest sales and profit; Central has the lowest profit ratio (7.9%).
======================================================================
STEP 3 : Worksheet 2 - Line chart: Sales by month and year
======================================================================
>> SUM(Sales) and SUM(Profit) per YEAR(Order Date):
          Sales    Profit  Sales growth %
Year
2015  484247.50  49543.97             NaN
2016  470532.51  61618.60           -2.83
2017  609205.60  81795.17           29.47
2018  733215.26  93439.27           20.36

>> Continuous MONTH(Order Date) series: 48 months; peaks and lows:
   Highest month : Nov 2018 ($118,447.82)
   Lowest month  : Feb 2015 ($4,519.89)

>> Sales by month (rows) and year (columns), in USD:
Year      2015     2016     2017      2018
Month
1      14237.0  18174.0  18542.0   43971.0
```

```
2       4520.0  11951.0  22979.0   20301.0
3      55691.0  38726.0  51716.0   58872.0
4      28295.0  34195.0  38750.0   36522.0
5      23648.0  30132.0  56988.0   44261.0
6      34595.0  24797.0  40345.0   52982.0
7      33946.0  28765.0  39262.0   45264.0
8      27909.0  36898.0  31115.0   63121.0
9      81777.0  64596.0  73410.0   87867.0
10     31453.0  31405.0  59688.0   77777.0
11     78629.0  75973.0  79412.0  118448.0
12     69546.0  74920.0  96999.0   83829.0
>> Plot saved to: plots/2_sheet_monthly_sales_line.png
>> Interpretation: sales dipped 2.8% in 2016, then grew strongly in 2017 and 2018.
   Clear seasonal pattern: low in Jan/Feb, highest in Sep, Nov and Dec.

======================================================================
STEP 4 : Worksheet 3 - Geographic (State map) and Segment (scatter) breakdowns
======================================================================
>> Filled map values: 49 states. Top 10 states by SUM(Sales):
                Sales    Profit
State
California    457687.63  76381.39
New York      310876.27  74038.55
Texas         170188.05 -25729.36
Washington    138641.27  33402.65
Pennsylvania  116511.91 -15559.96
Florida        89473.71  -3399.30
Illinois       80166.10 -12607.89
Ohio           78258.14 -16971.38
Michigan       76269.61  24463.19
Virginia       70636.72  18597.95

>> 5 states with the largest losses (shown red if Profit is on Color):
                  Sales    Profit
State
Texas           170188.05 -25729.36
Ohio             78258.14 -16971.38
Pennsylvania    116511.91 -15559.96
Illinois         80166.10 -12607.89
North Carolina   55603.16  -7490.91

>> Segment breakdown:
                Sales     Profit  Customers
Segment
Consumer     1161401.34  134119.21        409
Corporate     706146.37   91979.13        236
Home Office   429653.15   60298.68        148

>> Scatter plot: one mark per customer (793 marks), x = SUM(Sales), y = SUM(Profit), color = Segment
>> Plot saved to: plots/3_sheet_state_sales_top10.png
>> Plot saved to: plots/4_sheet_segment_scatter.png
>> Interpretation: California and New York lead sales; Texas loses the most money.
======================================================================
STEP 5 : Dashboard preview - all regions vs Region = 'West' selected
======================================================================
>> Plot saved to: plots/5_dashboard_all_regions.png

>> After clicking the 'West' bar (Filter action), the other sheets show:
   Order lines       : 3,203
   Total Sales       : $725,457.82
   Total Profit      : $108,418.45
   Customers in West : 686
   States in West    : 11
>> Plot saved to: plots/6_dashboard_filtered_west.png

======================================================================
STEP 6 : Lab Assignment 5 companion script completed
======================================================================
Files generated:
  - plots/1_sheet_region_bar.png
  - plots/2_sheet_monthly_sales_line.png
  - plots/3_sheet_state_sales_top10.png
  - plots/4_sheet_segment_scatter.png
  - plots/5_dashboard_all_regions.png
  - plots/6_dashboard_filtered_west.png
```